An Introduction to AR-EGARCH Models and Quasi-Maximum Likelihood Estimation*

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1 Introduction

Financial asset returns exhibit pronounced conditional heteroskedasticity, a property known as volatility clustering (Engle, 2001). Engle (2001) describes the rolling standard deviation as the first autoregressive conditional heteroskedasticity (ARCH) model, a restrictive specification that assigns equal weight to all observations within the window and zero weight to all earlier observations. The ARCH model introduced by Engle (1982) removed this restriction by letting the data determine the weights on lagged squared residuals, and Bollerslev (1986) extended it to the generalized ARCH (GARCH) model. To accommodate asymmetric leverage effects and remove the nonnegativity constraints on the parameters, Nelson (1991) formulated the exponential GARCH (EGARCH) model, which parameterizes the logarithm of the conditional variance so that return shocks affect volatility through both their magnitude and their sign.

Models of conditional heteroskedasticity provide the volatility measures used in risk assessment, asset allocation and contingent-claim valuation (Engle, 2001). A prominent application is value at risk (VaR), the loss that is exceeded only with a specified probability over a given holding period. For example, Engle (2001) constructs the VaR of a portfolio of stocks and bonds from the one-day-ahead conditional variance forecast of a GARCH(1,1) model. These models are available in standard software.[1]

A standard empirical specification of return dynamics comprises a conditional mean equation, typically an autoregressive model of order p, AR(p), and a conditional variance equation governed by a GARCH-type recursion. The parameters of both equations can be estimated jointly by maximum likelihood estimation (MLE), or sequentially, in which case the conditional mean is fitted first, and the conditional variance is then estimated from the resulting residuals. Both routes are used in practice. Nelson (1991) and Tsay (2010) estimate an AR(1) mean, augmented by an in-mean term in the former, and an EGARCH variance jointly, whereas Engle and Ng (1993) and Hafner and Linton (2017) fit EGARCH to the residuals of a first-step mean regression.

Sequential estimation offers computational advantages in applied work. It decomposes one high-dimensional optimization problem into lower-dimensional ones, and the second step can be carried out with standard econometric software on demeaned returns. When the model satisfies the symmetry and regularity conditions of Engle (1982) and the innovations are Gaussian, sequential estimation incurs no loss of asymptotic efficiency for the variance parameters relative to joint

*This note is intended for readers with a background in statistical inference and time series. It introduces the research question of the accompanying study, which examines the precision of parameter estimates in an autoregressive model with exponential GARCH errors.

[1]GARCH and EGARCH models are implemented in the R package rugarch, the Python package arch and the MATLAB Econometrics Toolbox.

estimation (Engle, 1982, Theorem 4). However, Bollerslev et al. (1994) note that this equivalence typically fails for asymmetric specifications such as EGARCH, in which the conditional variance depends on the sign of past shocks. Which parameters sequential estimation then estimates less precisely, and by how much, has not been determined. The accompanying study examines this question.

The remainder of this note is organized as follows. Section 2 reviews time series foundations and Section 3 documents the stylized facts of asset returns. Sections 4 and 5 present the GARCH and EGARCH models. Section 6 describes MLE and the quasi-maximum likelihood estimator (QMLE). Section 7 describes how the conditional volatility is filtered from observed returns. Section 8 compares joint and sequential estimation. Section 9 discusses the evaluation of volatility forecasts, and Section 10 concludes.

2 Time series foundations

2.1 Stationarity and ergodicity

A time series is a sequence of random variables $(y_t)_{t\in\mathbb{Z}}$, of which $y_1, \ldots, y_T$ are observed. Inference about its probabilistic properties typically relies on two assumptions. Stationarity requires the probabilistic structure to remain invariant over time, while ergodicity ensures that, under suitable integrability conditions, time averages converge almost surely to their corresponding expectations. Together, these properties provide a foundation for inferring population properties from a single realization.

Definition 2.1 (Stationarity)

The series (y_t) is strictly (strongly) stationary if the joint probability distribution of the process remains the same when shifted in time. It is covariance (weakly) stationary[a] if $\mathsf{E}[y_t^2] < \infty$ and

$$\mathsf{E}[y_t] = \mu \qquad \text{and} \qquad \mathsf{Cov}(y_t, y_{t-k}) = \lambda_k$$

do not depend on t.

[a] A strictly stationary series with finite variance is covariance stationary. The converse does not hold.

Ergodicity means that a single long realization eventually reflects the full distribution of the process. The practical content of ergodicity is the ergodic theorem. If (y_t) is strictly stationary and ergodic with $\mathsf{E}[\,|y_t|\,] < \infty$, then

$$\frac{1}{T}\sum_{t=1}^{T} y_t \xrightarrow{\text{a.s.}} \mathsf{E}[y_t]. \tag{2.1}$$

2.2 White noise, martingale differences and independence

Three notions of an unexpected shock are used throughout. Let $\mathcal{F}_{t-1}$ denote the information available at $t-1$. A sequence (ε_t) is

(i) white noise if $\mathsf{E}[\varepsilon_t] = 0$, $\mathsf{Var}(\varepsilon_t) = \sigma^2$ and $\mathsf{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$,

(ii) a martingale difference sequence if $\mathsf{E}[\,|\varepsilon_t|\,] < \infty$ and $\mathsf{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$,

(iii) independent and identically distributed (IID) with $\mathsf{E}[\varepsilon_t] = 0$.[2]

Volatility models exploit the gap between (ii) and (iii), since a return can be a martingale difference sequence whose square is autocorrelated. For such shocks, the martingale central limit theorem of Billingsley (1961)[3] underlies the asymptotic normality of the estimators.

2.3 Autoregressions

The AR(p) model is

$$y_t - \nu = \sum_{i=1}^{p} \phi_i (y_{t-i} - \nu) + \varepsilon_t, \tag{2.2}$$

where (ε_t) is a martingale difference sequence. With the lag operator L and the polynomial $\Phi(L) = 1 - \sum_{i=1}^{p} \phi_i L^i$, the model can be written as $\Phi(L)(y_t - \nu) = \varepsilon_t$.

If all roots of $\Phi(z)$ lie outside the unit circle and (ε_t) has constant finite variance, the model is covariance stationary and admits a moving average representation with absolutely summable coefficients (Hamilton, 1994)

$$y_t - \nu = \sum_{k \geq 0} \psi_k \varepsilon_{t-k}, \qquad \sum_{k \geq 0} \psi_k z^k = \Phi(z)^{-1}. \tag{2.3}$$

Example 2.1 (AR(1))

With $|\phi_1| < 1$ and $\mathsf{Var}(\varepsilon_t) = \sigma^2$, the variance and the autocorrelation at lag k are

$$\mathsf{Var}(y_t) = \frac{\sigma^2}{1 - \phi_1^2}, \qquad \varrho_k = \phi_1^k.$$

The effect of a shock therefore decays geometrically.

2.4 Least squares

Let $\boldsymbol{m} = (\nu, \boldsymbol{\phi}^T)^T$ denote the mean parameters, with $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_p)^T$. The residual is

$$\varepsilon_t(\boldsymbol{m}) = y_t - \nu - \sum_{i=1}^{p} \phi_i (y_{t-i} - \nu),$$

and the ordinary least squares (OLS) estimator $\hat{\boldsymbol{m}}_1$ minimizes $\sum_t \varepsilon_t(\boldsymbol{m})^2$, which amounts to regressing y_t on a constant and $y_{t-1}, \ldots, y_{t-p}$ with intercept $\nu\Phi(1)$. If (y_t) is strictly stationary and

[2] An IID sequence with finite variance is a martingale difference sequence, and a martingale difference sequence with constant finite unconditional variance is white noise. The converse implications do not hold.

[3] If (ε_t) is a strictly stationary and ergodic martingale difference sequence with $\mathsf{E}[\varepsilon_t^2] < \infty$, then $\frac{1}{\sqrt{T}} \sum_{t=1}^{T} \varepsilon_t \xrightarrow{\mathrm{d}} \mathrm{N}(0, \mathsf{E}[\varepsilon_t^2])$.

ergodic with sufficiently many finite moments,[4] then at the true value $\boldsymbol{m}_0$,

$$\begin{aligned}\sqrt{T}(\hat{\boldsymbol{m}}_1 - \boldsymbol{m}_0) &\xrightarrow{\mathrm{d}} \mathrm{N}(\mathbf{0}, V_{\mathrm{OLS}}),\\ V_{\mathrm{OLS}} &= M^{-1}\mathsf{E}[\varepsilon_t^2 \boldsymbol{\xi}_t \boldsymbol{\xi}_t^T] M^{-1}, \qquad M = \mathsf{E}[\boldsymbol{\xi}_t \boldsymbol{\xi}_t^T],\\ \boldsymbol{\xi}_t &= \frac{\partial \varepsilon_t(\boldsymbol{m}_0)}{\partial \boldsymbol{m}} = -\big(\Phi(1),\, y_{t-1} - \nu,\, \ldots,\, y_{t-p} - \nu\big)^T.\end{aligned} \tag{2.4}$$

If the conditional variance $\mathsf{E}[\varepsilon_t^2 \mid \mathcal{F}_{t-1}]$ is a constant σ^2, the law of iterated expectations gives $\mathsf{E}[\varepsilon_t^2 \boldsymbol{\xi}_t \boldsymbol{\xi}_t^T] = \sigma^2 M$ and hence $V_{\mathrm{OLS}} = \sigma^2 M^{-1}$. If the conditional variance depends on the past, $\mathsf{E}[\varepsilon_t^2 \boldsymbol{\xi}_t \boldsymbol{\xi}_t^T]$ in general differs from $\sigma^2 M$, so the heteroskedasticity-consistent form is required (Hamilton, 1994, Proposition 8.3).

If the conditional variance $\sigma_t^2 = \mathsf{E}[\varepsilon_t^2 \mid \mathcal{F}_{t-1}]$ were known, generalized least squares (GLS) would minimize $\sum_t \varepsilon_t(\boldsymbol{m})^2/\sigma_t^2$, weighting each observation by its precision. The inverse conditional variance is the optimal weight (Newey and McFadden, 1994), and the asymptotic covariance of GLS is $\mathsf{E}[\sigma_t^{-2} \boldsymbol{\xi}_t \boldsymbol{\xi}_t^T]^{-1}$, which is no larger than V_{OLS}. Feasible GLS (FGLS) replaces σ_t^2 by an estimate. For a regression whose error covariance matrix is a known function of parameters that are estimated consistently, this replacement often leaves the asymptotic distribution of GLS unchanged (Hamilton, 1994). That result does not cover lagged dependent variables as regressors, but the same conclusion holds for the AR(p)-EGARCH(1,1) model, because the expected derivative of the weighted moment with respect to the variance parameters is zero at the true value, which removes the first-step correction by the argument of Newey and McFadden (1994, Theorem 6.1), stated for independent data. The accompanying study assumes the resulting expansion.

Example 2.2 (Least squares in an AR(1))

Given ν, the OLS estimator has the closed form

$$\hat{\phi}_1 = \frac{\sum_{t=2}^{T} (y_t - \nu)(y_{t-1} - \nu)}{\sum_{t=2}^{T} (y_{t-1} - \nu)^2}.$$

Here $\boldsymbol{\xi}_t = -(y_{t-1} - \nu)$ and $M = \mathsf{Var}(y_t)$, so eq. (2.4) with IID errors yields

$$\sqrt{T}(\hat{\phi}_1 - \phi_1) \xrightarrow{\mathrm{d}} \mathrm{N}(0, V_{\mathrm{OLS}}), \qquad V_{\mathrm{OLS}} = \frac{\mathsf{E}[\varepsilon_t^2 (y_{t-1} - \nu)^2]}{\mathsf{Var}(y_t)^2} = \frac{\sigma^2}{\mathsf{Var}(y_t)} = 1 - \phi_1^2.$$

3 Stylized facts of asset returns

Empirical work has established a set of stylized facts (Cont, 2001),[5]

(i) Absence of autocorrelations [1]. The linear autocorrelations of asset returns are often insignif-

[4] Weaker conditions suffice. Hamilton (1994, Assumption 8.6) requires, besides moment conditions, that $\boldsymbol{\xi}_t \varepsilon_t$ be a martingale difference sequence and that the sample averages of $\boldsymbol{\xi}_t \boldsymbol{\xi}_t^T$ and $\varepsilon_t^2 \boldsymbol{\xi}_t \boldsymbol{\xi}_t^T$ converge in probability to positive definite limits. Weak stationarity alone does not ensure the latter, which requires a law of large numbers for these products.

[5] The six facts follow the statements of Cont (2001), whose numbering is given in brackets.

icant.[6]

(ii) Heavy tails [2]. The unconditional distribution of returns seems to display a power-law or Pareto-like tail.

(iii) Volatility clustering [6]. Different measures of volatility display a positive autocorrelation over several days, which quantifies the tendency of high-volatility events to cluster in time.

(iv) Slow decay of autocorrelation in absolute returns [8]. The autocorrelation function of absolute returns decays slowly with the lag, roughly as a power law.

(v) Leverage effect [9]. Most measures of the volatility of an asset are negatively correlated with the returns of that asset.

(vi) Conditional heavy tails [7]. Even after correcting for volatility clustering, the residuals still exhibit heavy tails, though less heavy than the unconditional distribution.

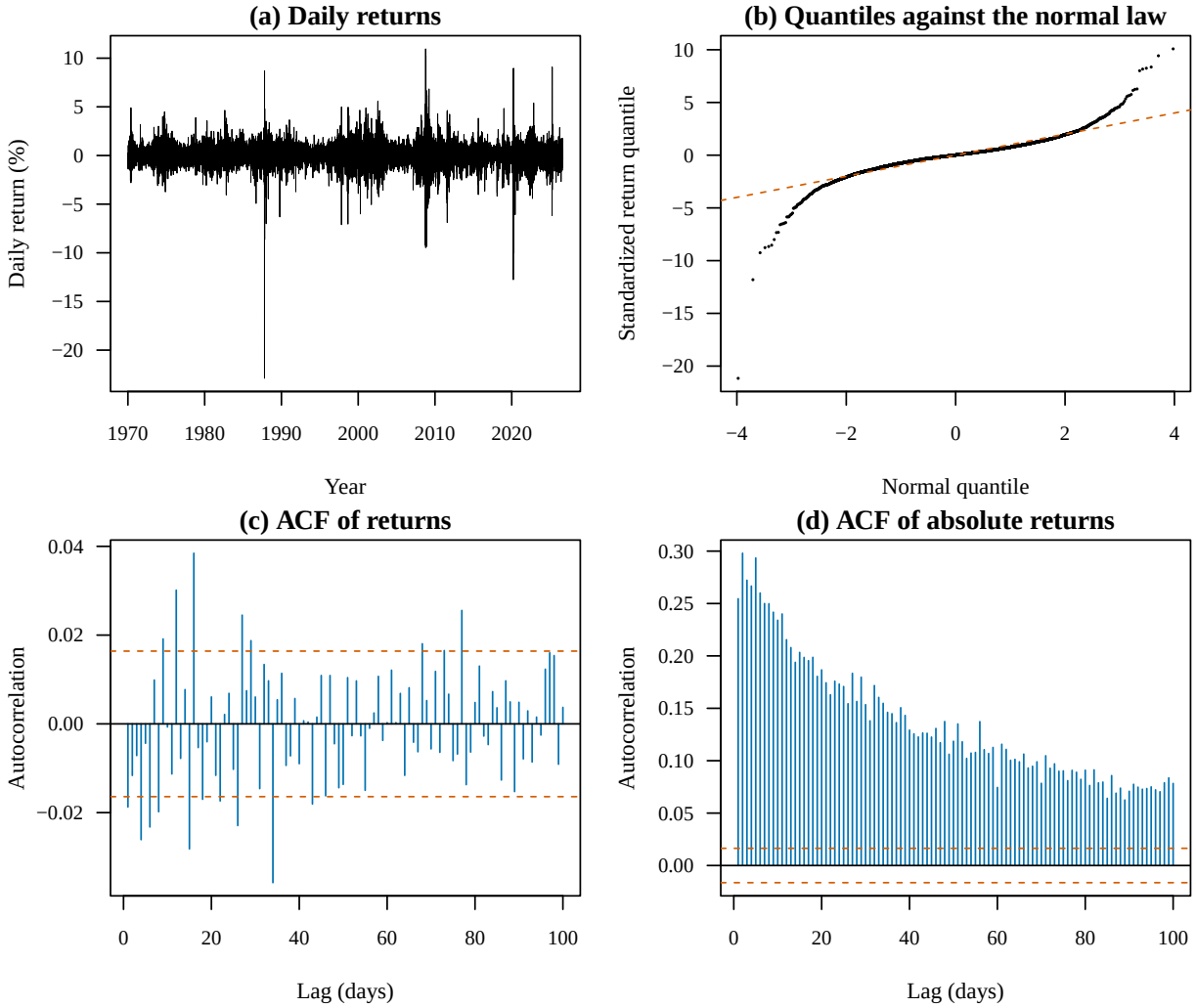


Figure 3.1: Stylized facts of daily S&P 500 log returns (%), January 1970 to August 2026.

[6]Except at intraday time scales of approximately 20 minutes, where microstructure effects come into play.

Figure 3.1 illustrates four of these facts in daily S&P 500 returns from January 1970 to August 2026. Panel (a) shows volatility clustering, with calm periods interrupted by volatile episodes such as 1987, 2008 and 2020. Panel (b) shows tails far heavier than those of the Gaussian law. Panels (c) and (d) show the sample autocorrelation functions (ACF). The sample autocorrelations of returns are small, at most 0.04 in absolute value, whereas those of absolute returns are all significant and decay slowly.

Example 3.1 (Volatility clustering creates heavy tails)

Let $\varepsilon_t = \sigma_t \eta_t$ with η_t independent of σ_t, $\mathsf{E}[\eta_t^2] = 1$ and raw kurtosis $\kappa_4 = \mathsf{E}[\eta_t^4]$. Then

$$\frac{\mathsf{E}[\varepsilon_t^4]}{(\mathsf{E}[\varepsilon_t^2])^2} = \kappa_4 \frac{\mathsf{E}[\sigma_t^4]}{(\mathsf{E}[\sigma_t^2])^2} \;\geq\; \kappa_4 \qquad \text{(by Jensen's inequality)},$$

with equality if and only if σ_t is constant. A time-varying volatility therefore generates unconditional heavy tails even under Gaussian innovations.

Each fact motivates a modeling choice. Fact (i) indicates that the conditional mean is nearly constant, so a low-order autoregression suffices. Facts (iii) and (iv) indicate that the conditional variance is predictable and persistent, which motivates a variance recursion. Fact (v) requires the variance to respond more to bad news than to good news. Fact (vi) indicates that the innovation itself is not Gaussian, so the Gaussian likelihood serves only as a quasi-likelihood in Section 6.

4 The GARCH model

A conditional volatility model writes the shock as

$$\varepsilon_t = \sigma_t \eta_t, \tag{4.1}$$

where (η_t) is IID with zero mean and unit variance, and η_t is independent of $\mathcal{F}_{t-1}$. Then $\varepsilon_t \mid \mathcal{F}_{t-1}$ has mean zero and variance σ_t^2. The GARCH(p, q) model of Bollerslev (1986)[7] is

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2, \qquad \omega > 0,\ \alpha_i \geq 0,\ \beta_j \geq 0, \tag{4.2}$$

where the nonnegativity constraints ensure that $\sigma_t^2 > 0$. This note focuses on GARCH(1,1), which predicts conditional variances well despite its parsimony (Engle, 2001),

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \qquad \omega > 0,\ \alpha \geq 0,\ \beta \geq 0. \tag{4.3}$$

The GARCH(1,1) process is covariance stationary if and only if $\alpha + \beta < 1$, in which case its long-run variance is $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$ (Bollerslev, 1986, Theorem 1). Substituting $\omega = (1 - \alpha - \beta)\bar{\sigma}^2$

[7] The ARCH(q) model of Engle (1982) is the case $p = 0$.

into eq. (4.3) gives

$$\sigma_t^2 = (1 - \alpha - \beta)\,\bar{\sigma}^2 + \beta\,\sigma_{t-1}^2 + \alpha\,\varepsilon_{t-1}^2. \tag{4.4}$$

The current variance is thus a weighted average of the long-run variance, the previous variance and the previous squared shock. The squared shocks follow an autoregressive moving average representation, which produces volatility clustering,

$$\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + \zeta_t - \beta\zeta_{t-1}, \qquad \zeta_t = \varepsilon_t^2 - \sigma_t^2. \tag{4.5}$$

Example 4.1 (Persistence)

Iterating eq. (4.4) and using $\mathsf{E}[\varepsilon_{t+k}^2 \mid \mathcal{F}_t] = \mathsf{E}[\sigma_{t+k}^2 \mid \mathcal{F}_t]$ gives the forecast

$$\mathsf{E}[\sigma_{t+k}^2 \mid \mathcal{F}_t] = \bar{\sigma}^2 + (\alpha + \beta)^{k-1}\big(\sigma_{t+1}^2 - \bar{\sigma}^2\big), \qquad k \geq 1,$$

which tends to $\bar{\sigma}^2$ as $k \to \infty$.

5 The EGARCH model

5.1 The model

Because GARCH models rule out by construction the negative correlation between current returns and future volatility, and their parameter constraints are often violated by estimated coefficients and may unduly restrict the dynamics, Nelson (1991) introduced EGARCH, which models the logarithm of the conditional variance $h_t = \log \sigma_t^2$,

$$h_t = \omega + \alpha\big(|\eta_{t-1}| - \mathsf{E}|\eta_{t-1}|\big) + \gamma\eta_{t-1} + \beta h_{t-1}, \tag{5.1}$$

where α is the size effect, γ is the sign effect, β measures the persistence of the log variance, and $\mathsf{E}|\eta_{t-1}|$ is a constant. Since the recursion is specified for the logarithm, σ_t^2 is positive for all parameter values, and no positivity constraint is needed.

5.2 News impact and leverage

Engle and Ng (1993) summarize the response of the variance to news by the news impact curve. It relates σ_t^2 to ε_{t-1} with the lagged variance at its unconditional level, and for GARCH(1,1) the curve is

$$\sigma_t^2 = \omega + \beta\bar{\sigma}^2 + \alpha\varepsilon_{t-1}^2,$$

a parabola symmetric about $\varepsilon_{t-1} = 0$. A fall and a rise of equal size raise the variance by the same amount, so GARCH models cannot generate the leverage effect, fact (v). In EGARCH, the news

term of eq. (5.1) is piecewise linear in the standardized shock $\eta = \eta_{t-1}$,

$$\alpha(|\eta| - \mathsf{E}|\eta|) + \gamma\eta = \begin{cases} (\alpha+\gamma)\,\eta - \alpha\mathsf{E}|\eta|, & \eta \geq 0, \\ (\gamma-\alpha)\,\eta - \alpha\mathsf{E}|\eta|, & \eta < 0. \end{cases} \tag{5.2}$$

The log variance therefore increases with the size of the shock when $\alpha > |\gamma|$, and increases more after a fall when $\gamma < 0$. Since the news enters the logarithm, the news impact curve of σ_t^2 is exponential on each side of zero rather than quadratic, with different slopes on the two sides when $\gamma \neq 0$ (Engle and Ng, 1993).

5.3 Stationarity and moments

Iterating eq. (5.1) yields a moving average representation,

$$h_t = \frac{\omega}{1-\beta} + \sum_{j\geq 1} \beta^{j-1}\big\{\alpha(|\eta_{t-j}| - \mathsf{E}|\eta_{t-j}|) + \gamma\eta_{t-j}\big\}.$$

Nelson (1991, Theorem 2.1) shows that the log variance is strictly stationary and ergodic if and only if the squared coefficients are summable, which for the first-order model means $|\beta| < 1$ when α and γ are not both zero. The existence of its unconditional moments depends on the exponential integrability of the innovation distribution because σ_t^2 is the exponential of a sum of terms linear in $|\eta|$.[8] The shock $\varepsilon_t = e^{h_t/2}\eta_t$ is a measurable function of $(\eta_s)_{s\leq t}$, and so is the AR(p) process. Both are therefore strictly stationary and ergodic (Straumann and Mikosch, 2006, Proposition 2.5).

6 Likelihood and quasi-maximum likelihood

6.1 The Gaussian likelihood

The parameters of the AR(p)-EGARCH(1,1) model are $\boldsymbol{\theta} = (\boldsymbol{m}^T, \boldsymbol{v}^T)^T$, with mean part $\boldsymbol{m}$ and variance part $\boldsymbol{v} = (\omega, \alpha, \gamma, \beta)^T$. If η_t were Gaussian, the log density of observation t would be, up to a constant,

$$l_t(\boldsymbol{\theta}) = -\frac{1}{2}\Big\{h_t(\boldsymbol{\theta}) + \varepsilon_t(\boldsymbol{m})^2 e^{-h_t(\boldsymbol{\theta})}\Big\}. \tag{6.1}$$

Here, $h_t(\boldsymbol{\theta})$ denotes the log variance at $\boldsymbol{\theta}$. The first term penalizes a large variance, and the second penalizes a residual that is large relative to the variance.

6.2 Score and information matrices

Two matrices govern the precision of the maximizer $\hat{\boldsymbol{\theta}}$ of $\sum_t l_t(\boldsymbol{\theta})$. The score is $\boldsymbol{s}_t = \partial l_t(\boldsymbol{\theta}_0)/\partial\boldsymbol{\theta}$, and

$$\mathcal{I} = \mathsf{E}[\boldsymbol{s}_t \boldsymbol{s}_t^T], \qquad \mathcal{J} = -\mathsf{E}\Big[\frac{\partial^2 l_t(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}^T}\Big]. \tag{6.2}$$

[8] For the generalized error distribution with shape parameter above one, which includes the Gaussian law, all moments of σ_t^2 are finite (Nelson, 1991, Theorem 2.2). With Student t innovations, by contrast, the conditional variance typically has no finite unconditional moments (Nelson, 1991).

The matrix $\mathcal{I}$ measures the variability of the score, and $\mathcal{J}$ measures the curvature of the expected log-likelihood at the true parameter. If (y_t) is strictly stationary and ergodic, then under regularity conditions the estimator is consistent and asymptotically normal,[9]

$$\sqrt{T}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{\mathrm{d}} \mathrm{N}\big(\mathbf{0},\ \mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}\big). \tag{6.3}$$

For IID data from a correctly specified model, the information matrix equality $\mathcal{I} = \mathcal{J}$ holds (White, 1982, Theorem 3.3), and the asymptotic covariance reduces to $\mathcal{I}^{-1}$, the lower bound for the asymptotic covariance of generalized method of moments estimators (Newey and McFadden, 1994, Theorem 5.1).

6.3 Quasi-maximum likelihood

Returns are not conditionally Gaussian (fact (vi)), so eq. (6.1) is a quasi-likelihood, and its maximizer is the Gaussian QMLE. The estimator remains consistent because only the first two conditional moments enter its score. The score is

$$\boldsymbol{s}_t = \underbrace{-\frac{1}{2}\big(1 - \eta_t^2\big)\frac{\partial h_t}{\partial \boldsymbol{\theta}}}_{\text{variance channel}} \underbrace{- \eta_t e^{-h_t/2}\frac{\partial \varepsilon_t}{\partial \boldsymbol{\theta}}}_{\text{mean channel}}. \tag{6.4}$$

Since $\mathsf{E}[\eta_t] = 0$ and $\mathsf{E}[\eta_t^2] = 1$, the score satisfies $\mathsf{E}[\boldsymbol{s}_t \mid \mathcal{F}_{t-1}] = \mathbf{0}$ whatever the law of η_t, so $(\boldsymbol{s}_t)$ is a martingale difference sequence at $\boldsymbol{\theta}_0$. The expected quasi-likelihood is maximized at $\boldsymbol{\theta}_0$, uniquely under identification, and eq. (6.3) holds with $\mathcal{I} \neq \mathcal{J}$ in general. For IID data, White (1982, Theorems 2.2 and 3.2) establishes consistency for the pseudo-true value, which minimizes the Kullback–Leibler divergence from the true law, and this sandwich form. Bollerslev and Wooldridge (1992, Theorem 2.1) establish the sandwich form for dynamic models with correctly specified conditional mean and variance.

The Gaussian likelihood can therefore serve as a valid quasi-likelihood estimating criterion for the conditional mean and variance under suitable regularity and identification conditions, even when the innovation distribution is non-Gaussian.

Example 6.1 (Estimating a variance)

Let ε_t be IID with mean zero, variance σ^2 and raw kurtosis κ_4, and let $\hat{\sigma}^2 = T^{-1}\sum_t \varepsilon_t^2$ be the Gaussian QMLE of σ^2. Then

$$\mathsf{Var}(\hat{\sigma}^2) = \frac{\mathsf{Var}(\varepsilon_t^2)}{T} = \frac{(\kappa_4 - 1)\sigma^4}{T}.$$

The same factor appears in volatility models. When only the variance parameters are esti-

[9] Newey and McFadden (1994, Theorem 2.1) establish consistency from identification, compactness of the parameter set, continuity of the limit objective and uniform convergence in probability of the sample objective to it. Newey and McFadden (1994, Theorem 3.1) establish asymptotic normality from consistency, an interior true value, twice continuous differentiability near it, a central limit theorem for the score, uniform convergence in probability of the Hessian near the true value to a limit continuous there, and nonsingularity of $\mathcal{J}$.

mated, the mean channel of eq. (6.4) drops out and

$$\mathcal{I}_{\boldsymbol{vv}} = \frac{\kappa_4 - 1}{4}\,\mathsf{E}\Big[\frac{\partial h_t}{\partial \boldsymbol{v}}\frac{\partial h_t}{\partial \boldsymbol{v}^T}\Big], \qquad \mathcal{J}_{\boldsymbol{vv}} = \frac{1}{2}\,\mathsf{E}\Big[\frac{\partial h_t}{\partial \boldsymbol{v}}\frac{\partial h_t}{\partial \boldsymbol{v}^T}\Big],$$
$$\mathcal{J}_{\boldsymbol{vv}}^{-1}\mathcal{I}_{\boldsymbol{vv}}\mathcal{J}_{\boldsymbol{vv}}^{-1} = (\kappa_4 - 1)\Big\{\mathsf{E}\Big[\frac{\partial h_t}{\partial \boldsymbol{v}}\frac{\partial h_t}{\partial \boldsymbol{v}^T}\Big]\Big\}^{-1}.$$

7 Filtering the volatility and invertibility

7.1 The filter

To evaluate the log density eq. (6.1) at $\boldsymbol{\theta}$, the log variance is computed from the data by running the recursion forward from a starting value $\hat{h}_0$,

$$\begin{aligned} \hat{h}_t(\boldsymbol{\theta}) &= \mathcal{T}_{t-1}^{\boldsymbol{\theta}}\big(\hat{h}_{t-1}(\boldsymbol{\theta})\big), \\ \mathcal{T}_s^{\boldsymbol{\theta}}(x) &= \omega + \alpha\big(|\varepsilon_s(\boldsymbol{m})|e^{-x/2} - \mathsf{E}|\eta_s|\big) + \gamma\,\varepsilon_s(\boldsymbol{m})e^{-x/2} + \beta x. \end{aligned} \tag{7.1}$$

The filter[10] is invertible at the true parameter if the effect of the initialization vanishes asymptotically, so that $\hat{h}_t(\boldsymbol{\theta}_0) - h_t \to 0$ in probability from any starting value (Straumann and Mikosch, 2006). It is continuously invertible if $\sup_{\boldsymbol{\theta}} |\hat{h}_t(\boldsymbol{\theta}) - h_t(\boldsymbol{\theta})| \to 0$ almost surely, the supremum running over the parameter set (Wintenberger, 2013, Definition 2). Invertibility (Straumann and Mikosch, 2006) and its uniform version (Wintenberger, 2013) play an important role in establishing consistency and asymptotic normality of the EGARCH QMLE. Without suitable invertibility conditions, these asymptotic results of Section 6 are not guaranteed.[11]

7.2 Invertibility of the GARCH and EGARCH filters

For GARCH(1,1), the data enter two runs of the filter identically, and the recursion is linear in the lagged variance with coefficient β. Applied to a run started at the true variance, this gives $\hat{\sigma}_t^2 - \sigma_t^2 = \beta^t(\hat{\sigma}_0^2 - \sigma_0^2) \to 0$, so the GARCH filter is invertible whenever $0 \le \beta < 1$ (Wintenberger, 2013).

The EGARCH map in eq. (7.1) is nonlinear, with slope

$$\frac{\mathrm{d}\mathcal{T}_s^{\boldsymbol{\theta}}(x)}{\mathrm{d}x} = \beta - \frac{1}{2}\big\{\alpha|\varepsilon_s(\boldsymbol{m})| + \gamma\,\varepsilon_s(\boldsymbol{m})\big\}e^{-x/2}. \tag{7.2}$$

When the filtered log variance x lies far below the true log variance h_s, the standardized residual is large and the slope can exceed one in absolute value. The discrepancy between two runs then grows in that step instead of shrinking, but an expanding step does not by itself destroy invertibility. The filter is a stochastic recurrence equation with random maps $\mathcal{T}_s$, which depend on the data

[10] The map uses the standardized residual $\varepsilon_s(\boldsymbol{m})e^{-x/2}$, because the log variance is x. In practice, $\mathsf{E}|\eta_s|$ is unknown and is set to its Gaussian value $\sqrt{2/\pi}$. Another value leaves the filter unchanged once ω is shifted.

[11] Invertibility also matters for forecasting, since a forecast that depends on an arbitrary starting value is unreliable (Wintenberger, 2013).

(Straumann and Mikosch, 2006). Each maps the interval $I = [(\omega - \alpha \mathsf{E}|\eta_s|)/(1-\beta), \infty)$ into itself when $0 \le \beta < 1$ and $\alpha \ge |\gamma|$. On I, each map is Lipschitz with random coefficient $\Lambda(\mathcal{T}_s) = \sup_{x \in I} |\mathrm{d}\mathcal{T}_s(x)/\mathrm{d}x|$. Suppose that $(\mathcal{T}_s)$ is stationary and ergodic, that $\mathsf{E}[\log^+ |\mathcal{T}_0(y) - y|]$ for some y and $\mathsf{E}[\log^+ \Lambda(\mathcal{T}_0)]$ are finite, and that the r-fold iterate of the maps is contractive on average,

$$\mathsf{E}\big[\log \Lambda\big(\mathcal{T}_s \circ \cdots \circ \mathcal{T}_{s-r+1}\big)\big] < 0. \tag{7.3}$$

Single steps may then expand, but over r steps the product of slopes shrinks on average on the log scale. The recursion then has a stationary solution, and the effect of the starting value vanishes exponentially fast almost surely (Bougerol, 1993, Theorem 3.1).[12]

At the true parameter, these conditions form INV($\boldsymbol{\theta}_0$) of Wintenberger (2013). For $r = 1$, Straumann and Mikosch (2006, Proposition 3.9) give eq. (7.3) as a sufficient condition for the invertibility of EGARCH(1,1). The expectation runs over infinitely many past innovations and has no closed form, so it is evaluated by simulation (Straumann and Mikosch, 2006). Wintenberger (2013) introduces continuous invertibility and the stable QMLE, which maximizes the quasi-likelihood over an empirical region in which $\alpha \ge |\gamma|$ and the sum of the log Lipschitz coefficients along the sample is bounded by a negative constant.

8 Joint and sequential estimation

8.1 Three estimators

For the AR(p)-EGARCH(1,1) model, three estimators of $\boldsymbol{\theta} = (\boldsymbol{m}^T, \boldsymbol{v}^T)^T$ are considered.

- The joint estimator (J) maximizes $\sum_t l_t(\boldsymbol{m}, \boldsymbol{v})$.
- The two-step estimator (2S) fits the mean by OLS, giving $\hat{\boldsymbol{m}}_1$, and then maximizes $\sum_t l_t(\hat{\boldsymbol{m}}_1, \boldsymbol{v})$ over $\boldsymbol{v}$ alone.
- The three-step estimator (3S) refits the mean by FGLS with the fitted variances, minimizing $\sum_t \varepsilon_t(\boldsymbol{m})^2 e^{-\hat{h}_t}$, where $\hat{h}_t$ is the log variance at the 2S estimate.

The 2S and 3S estimators follow the procedure of Engle (1982), except for the third step. Engle's third step is one scoring step for the regression parameters, which yields their efficient estimate given efficient estimates of the variance parameters. Unlike GLS, the scoring step accounts for the dependence of future conditional variances on the regression parameters. A one-step estimator, one Newton–Raphson iteration on all parameters from a $\sqrt{T}$-consistent start, has the same asymptotic variance as the maximizer, with or without block diagonality, under the conditions of Newey and McFadden (1994, Theorem 3.5), which include a twice continuously differentiable objective, and Engle's step, which updates only the regression parameters, does so for the mean under block diagonality. Ling and Zhu (2022) apply a one-step local QMLE for ARMA models with GARCH errors to all parameters after a self-weighted least squares estimate of the mean and a residual-based QMLE of the variance. These third steps aim at the efficiency of the J estimator.

[12] Straumann and Mikosch (2006, Theorem 2.8) restate this result and add that the stationary solution is unique.

The third step of the 3S estimator serves a different purpose. Let $W = \mathsf{E}[\sigma_t^{-2}\boldsymbol{\xi}_t\boldsymbol{\xi}_t^T]$, so that GLS has asymptotic covariance W^{-1}, and let $V_{\mathrm{J},\boldsymbol{mm}}$ be the mean block of $\mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}$. The efficiency gap between OLS and the J estimator of the mean splits into two parts,

$$V_{\mathrm{OLS}} - V_{\mathrm{J},\boldsymbol{mm}} = \big(V_{\mathrm{OLS}} - W^{-1}\big) + \big(W^{-1} - V_{\mathrm{J},\boldsymbol{mm}}\big).$$

The first part is the gain from weighting each observation by its conditional variance. The second is the difference due to also using the variance equation, in which the mean parameters enter h_t through the residuals. A third step that reaches the J estimator combines the two parts. FGLS attains W^{-1} without correcting for the estimated weights, so each part is the gain of a feasible estimator, where the first is from OLS to 3S, and the second is from 3S to J.[13]

Sequential estimation is attractive for three reasons. OLS requires no model for the conditional variance and is consistent and asymptotically normal under conditional heteroskedasticity of unknown form. The second step fits the pure EGARCH model, whose QMLE theory applies when the residuals are observed rather than estimated, with consistency by Straumann and Mikosch (2006, Theorem 5.1) and asymptotic normality of the stable QMLE by Wintenberger (2013, Theorem 6). MLE of these models is numerically demanding. Analytic derivatives in ARCH models "often involve very complicated recursive expressions", so that applications commonly rely on numerical derivatives (Bollerslev et al., 1994), and MLE of EGARCH "may be highly sensitive to starting values, the choice of numerical optimization algorithm, etc." (Hafner and Linton, 2017).

The OLS step has two drawbacks. First, OLS ignores the conditional variance, and its loss of efficiency relative to MLE "may be arbitrarily large" (Bollerslev et al., 1994). Second, the OLS estimation error can enter the second step, as derived next.

8.2 The cost of a first-step estimate

Partition $\boldsymbol{\theta} = (\boldsymbol{m}^T, \boldsymbol{v}^T)^T$, and $\mathcal{J}$ and $\mathcal{I}$ conformably into the blocks $\mathcal{J}_{\boldsymbol{mm}}$, $\mathcal{J}_{\boldsymbol{mv}}$, $\mathcal{J}_{\boldsymbol{vm}}$, $\mathcal{J}_{\boldsymbol{vv}}$ and likewise for $\mathcal{I}$. The cross blocks are

$$\mathcal{J}_{\boldsymbol{vm}} = -\mathsf{E}\Big[\frac{\partial^2 l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{v}\,\partial \boldsymbol{m}^T}\Big], \qquad \mathcal{I}_{\boldsymbol{vm}} = \mathsf{E}\big[\boldsymbol{s}_t^{(\boldsymbol{v})}\boldsymbol{s}_t^{(\boldsymbol{m})T}\big],$$

where $\boldsymbol{s}_t^{(\boldsymbol{m})}$ and $\boldsymbol{s}_t^{(\boldsymbol{v})}$ are the mean and variance blocks of the score. The cross block $\mathcal{J}_{\boldsymbol{vm}}$ measures how the variance score responds to the mean parameter, and $\mathcal{I}_{\boldsymbol{vm}}$ is the covariance between the variance and mean scores.

The OLS estimation error is an average of contributions from individual observations,

$$\sqrt{T}(\hat{\boldsymbol{m}}_1 - \boldsymbol{m}_0) = \frac{1}{\sqrt{T}}\sum_{t=1}^{T}\boldsymbol{p}_t + o_p(1), \qquad \boldsymbol{p}_t = -M^{-1}\boldsymbol{\xi}_t\varepsilon_t,$$

where $\boldsymbol{p}_t$ is the influence function of OLS. The second step of 2S solves $\sum_t \partial l_t(\hat{\boldsymbol{m}}_1, \boldsymbol{v})/\partial \boldsymbol{v} = \mathbf{0}$ with

[13] The covariance W^{-1} contains no factor $\kappa_4 - 1$, whereas the second part does, and the accompanying study shows that it can be negative when the tails are heavy. The losses studied are those of the 2S and 3S estimators.

the estimated $\hat{\boldsymbol{m}}_1$, and a first-order expansion in both arguments gives

$$\begin{aligned}\sqrt{T}(\hat{\boldsymbol{v}}_2 - \boldsymbol{v}_0) &= \mathcal{J}_{\boldsymbol{vv}}^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \boldsymbol{s}_t^{(v)} - \mathcal{J}_{\boldsymbol{vv}}^{-1} \mathcal{J}_{\boldsymbol{vm}} \sqrt{T}(\hat{\boldsymbol{m}}_1 - \boldsymbol{m}_0) + o_p(1) \\ &= \mathcal{J}_{\boldsymbol{vv}}^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{T} \left(\boldsymbol{s}_t^{(v)} - \mathcal{J}_{\boldsymbol{vm}} \boldsymbol{p}_t\right) + o_p(1).\end{aligned} \tag{8.1}$$

The first-step error enters the second step through $\mathcal{J}_{\boldsymbol{vm}}$, and ignoring it yields the uncorrected covariance $\mathcal{J}_{\boldsymbol{vv}}^{-1}\mathcal{I}_{\boldsymbol{vv}}\mathcal{J}_{\boldsymbol{vv}}^{-1}$. This is a special case of the general theory of two-step estimators. Newey (1984) shows that sequential estimators are method of moments estimators of the stacked moment conditions of both steps.[14]

8.3 Block diagonality

If $\mathcal{J}_{\boldsymbol{vm}} = 0$, the matrix $\mathcal{J}$ is block diagonal and the correction term in eq. (8.1) vanishes. The 2S and J estimators of $\boldsymbol{v}$ then have the same asymptotic covariance $\mathcal{J}_{\boldsymbol{vv}}^{-1}\mathcal{I}_{\boldsymbol{vv}}\mathcal{J}_{\boldsymbol{vv}}^{-1}$, for any $\mathcal{I}_{\boldsymbol{vm}}$, and the uncorrected standard errors are valid.

Engle (1982, Theorem 4) proves block diagonality for the ARCH regression model when it is symmetric and regular, and the innovations are Gaussian. Symmetry requires the conditional variance and its derivatives with respect to the variance parameters to be even functions of the lagged residuals. Regularity requires the conditional variance to be bounded below by a positive constant and the conditional expectations of products of its derivatives with respect to the variance parameters and the lagged residuals to exist. Bollerslev (1986) states that the off-diagonal block is zero for the GARCH regression model. For ARMA models with GARCH errors and symmetric innovations, block diagonality is part of the asymptotic theory of the joint QMLE of Francq and Zakoïan (2004, Theorem 3.2) and Ling and McAleer (2003).

Example 8.1 (Why symmetry gives block diagonality)

Let $y_t = \nu + \varepsilon_t$ with ARCH(1) errors $\sigma_t^2 = \omega + \alpha(y_{t-1} - \nu)^2$ and a symmetric innovation law, so that $\mu_3 = \mathsf{E}[\eta_t^3] = 0$. Write $u_t = \eta_t^2 - 1$. The scores for ν and α are

$$s_{\nu,t} = \frac{\varepsilon_t}{\sigma_t^2} - \alpha u_t \frac{\varepsilon_{t-1}}{\sigma_t^2}, \qquad s_{\alpha,t} = \frac{1}{2} u_t \frac{\varepsilon_{t-1}^2}{\sigma_t^2}.$$

Conditioning on $\mathcal{F}_{t-1}$ gives

$$\begin{aligned}\mathcal{J}_{\nu\alpha} &= -\alpha\, \mathsf{E}\left[\frac{\varepsilon_{t-1}^3}{\sigma_t^4}\right], \\ \mathcal{I}_{\nu\alpha} &= \frac{\mu_3}{2}\, \mathsf{E}\left[\frac{\varepsilon_{t-1}^2}{\sigma_t^3}\right] - \frac{\alpha(\kappa_4 - 1)}{2}\, \mathsf{E}\left[\frac{\varepsilon_{t-1}^3}{\sigma_t^4}\right].\end{aligned}$$

[14] Newey and McFadden (1994, Theorem 6.1) give the asymptotic variance with the correction for the first step.

The first term of $\mathcal{I}_{\nu\alpha}$ vanishes because $\mu_3 = 0$. The remaining integrand is a function of $\eta = \eta_{t-1}$,

$$f(\eta) = \frac{\varepsilon_{t-1}^3}{\sigma_t^4} = \frac{\sigma_{t-1}^3 \eta^3}{(\omega + \alpha\sigma_{t-1}^2\eta^2)^2}, \qquad f(-\eta) = -f(\eta).$$

Under a symmetric law the contributions at η and $-\eta$ cancel, so the conditional expectation is zero, and by the law of iterated expectations $\mathcal{J}_{\nu\alpha} = \mathcal{I}_{\nu\alpha} = 0$.

In EGARCH, γ multiplies an odd function of the shock, so the model lacks the symmetry that guarantees block diagonality, even when its variance responds symmetrically. Example 8.1 has an odd derivative with respect to the mean and an even derivative with respect to the variance parameter,

$$\frac{\partial \sigma_t^2}{\partial \nu} = -2\alpha\varepsilon_{t-1} \quad \text{(odd)}, \qquad \frac{\partial \sigma_t^2}{\partial \alpha} = \varepsilon_{t-1}^2 \quad \text{(even)},$$

so their product is odd and has mean zero under a symmetric law (Francq and Zakoïan, 2004, Lemma 4.1). In EGARCH at $\gamma_0 = 0$, the log variance is even in each lagged residual, but the derivative of the filter with respect to γ satisfies

$$\frac{\partial h_t}{\partial \gamma} = \eta_{t-1} + A_{t-1}\frac{\partial h_{t-1}}{\partial \gamma} \quad \text{(odd)}, \qquad A_{t-1} = \beta - \tfrac{1}{2}\alpha|\eta_{t-1}| \quad \text{(even)},$$

so it changes sign when the sign of every innovation is reversed, and the symmetry condition of Engle (1982) fails. The derivative with respect to the mean intercept is also odd, so their product contains $|\eta_{t-1}|$, which is even and has a positive mean, and parity no longer forces the cross term to vanish. Parity is sufficient for block diagonality but not necessary, so whether the cross block is nonzero must be determined directly. Bollerslev et al. (1994) attribute the failure of block diagonality for EGARCH to the asymmetric response of the variance, which applies when $\gamma_0 \neq 0$, whereas the argument above applies even when the response is symmetric.

9 Volatility forecast evaluation

The conditional variance is unobservable, so a forecast $\hat{\sigma}_{t+1}^2$ must be evaluated against a proxy. The squared shock is conditionally unbiased but noisy,

$$\mathsf{E}[\varepsilon_{t+1}^2 \mid \mathcal{F}_t] = \sigma_{t+1}^2, \qquad \mathsf{Var}(\varepsilon_{t+1}^2 \mid \mathcal{F}_t) = (\kappa_4 - 1)\sigma_{t+1}^4.$$

Andersen and Bollerslev (1998) show, for the GARCH(1,1) model, that this noise bounds the R^2 of a regression of squared returns even on the true variance by $1/\kappa_4$, one third under Gaussian innovations, so a low R^2 does not indicate a poor forecast. Patton (2011) asks which loss functions rank forecasts identically with a conditionally unbiased proxy $\tilde{\sigma}_{t+1}^2$ and with the true variance. The mean squared error (MSE) loss and the quasi-likelihood (QLIKE) loss,

$$\text{MSE} = \left(\tilde{\sigma}_{t+1}^2 - \hat{\sigma}_{t+1}^2\right)^2, \qquad \text{QLIKE} = \log \hat{\sigma}_{t+1}^2 + \frac{\tilde{\sigma}_{t+1}^2}{\hat{\sigma}_{t+1}^2}, \tag{9.1}$$

are both robust in this sense (Patton, 2011, Proposition 2). QLIKE equals -2 times the Gaussian log quasi-likelihood in eq. (6.1) with $\tilde{\sigma}^2_{t+1}$ in place of ε^2_{t+1}. With $\tilde{\sigma}^2_{t+1} = \varepsilon^2_{t+1}$, its conditional expectation for a forecast $f > 0$ is $\log f + \sigma^2_{t+1}/f$, which is minimized at $f = \sigma^2_{t+1}$ however noisy the proxy.

10 Conclusion

Asset returns are generally close to unpredictable in their mean but not in their variance. Their squares are autocorrelated, tails are heavy, and volatility rises more after falls than after rises (Cont, 2001). Models of conditional heteroskedasticity describe this dependence through a recursion for the conditional variance, from the ARCH model of Engle (1982) and the GARCH model of Bollerslev (1986) to the EGARCH model of Nelson (1991). EGARCH specifies the recursion for the log variance, which removes the positivity constraints and allows the variance to respond to the sign of a shock through γ.

Since the innovations are not Gaussian, these models are estimated by Gaussian QMLE. Its score depends only on the first two conditional moments and is a martingale difference sequence whatever the innovation law, so under identification the estimator is consistent and asymptotically normal with the sandwich covariance $\mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}$ under regularity conditions that include a twice continuously differentiable conditional mean and variance (Bollerslev and Wooldridge, 1992, Theorem 2.1). The standard asymptotic theory behind these results relies on the stationarity and ergodicity assumptions, and uses the martingale central limit theorem of Billingsley (1961). For EGARCH, it also requires the filter that recovers the latent log variance from the returns to be invertible, which holds when the r-fold iterate of its maps is contractive on average, under logarithmic moment conditions (Bougerol, 1993, Theorem 3.1). The same quasi-likelihood reappears in forecast evaluation, where its loss, QLIKE, ranks forecasts as the true variance would even when only a conditionally unbiased proxy is observed (Patton, 2011).

In applied work, the mean and the variance are often estimated in sequence, by OLS and then by QMLE on the residuals, because this lowers the dimension of each optimization. OLS remains consistent under conditional heteroskedasticity of unknown form, but its estimation error enters the second step through the cross block $\mathcal{J}_{\boldsymbol{vm}}$, so sequential estimation of the variance parameters is as efficient as joint estimation when the information matrix is block diagonal. Block diagonality holds for symmetric and regular ARCH regression models with Gaussian innovations (Engle, 1982, Theorem 4) and for ARMA models with GARCH errors and symmetric innovations (Francq and Zakoïan, 2004, Theorem 3.2). In both cases, the derivative of the variance with respect to the mean is odd in the lagged residuals while its derivatives with respect to the variance parameters are even. EGARCH breaks this parity even without leverage, since γ multiplies an odd function of the shock, and Bollerslev et al. (1994) note that block diagonality typically fails for it. Which entries of the cross block are nonzero, and how much precision each parameter then loses, has not been determined.

The asymptotic theory of EGARCH is also incomplete. Straumann and Mikosch (2006, Theorem 5.1) prove consistency of the QMLE for EGARCH(1,1), and Wintenberger (2013, Theorem 6)

proves asymptotic normality of the stable QMLE for the pure model under invertibility and moment conditions. Hafner and Kyriakopoulou (2021) prove asymptotic normality of the joint QMLE for EGARCH-in-mean models, but whether the argument extends to an autoregressive mean has not been investigated. Meitz and Saikkonen (2011, Example 2) obtain only consistency for a linear AR(p) model combined with an asymmetric GARCH model, because the absolute value in that model violates the twice differentiability assumption of the general theory, and its Lipschitz assumption rules out EGARCH (Meitz and Saikkonen, 2011, Assumption C2).[15] When introducing EGARCH, Nelson (1991) noted that "a satisfactory asymptotic theory for exponential ARCH is as yet unavailable" and assumed the MLE to be consistent and asymptotically normal. The accompanying study likewise imposes the asymptotic expansions of the estimators as a high-level assumption and examines the following question.

> In the AR(p)-EGARCH(1,1) model with a symmetric innovation that has exponential moments, which parameters does sequential estimation (OLS followed by QMLE, or an FGLS refit) estimate less efficiently than joint Gaussian QMLE?

References

Andersen, T. G., and T. Bollerslev. 1998. Answering the Skeptics: Yes, Standard Volatility Models Do Provide Accurate Forecasts. *International Economic Review* 39: 885–905.

Arvanitis, S., and S. Anyfantaki. 2020. On the Limit Theory of the Gaussian SQMLE in the EGARCH(1,1) Model. *Journal of Time Series Analysis* 41: 341–350.

Billingsley, P. 1961. The Lindeberg–Lévy Theorem for Martingales. *Proceedings of the American Mathematical Society* 12: 788–792.

Bollerslev, T. 1986. Generalized Autoregressive Conditional Heteroskedasticity. *Journal of Econometrics* 31: 307–327.

Bollerslev, T., R. F. Engle, and D. B. Nelson. 1994. ARCH Models. In R. F. Engle and D. L. McFadden (eds.), *Handbook of Econometrics*, vol. 4, 2959–3038. Amsterdam: Elsevier.

Bollerslev, T., and J. M. Wooldridge. 1992. Quasi-Maximum Likelihood Estimation and Inference in Dynamic Models with Time-Varying Covariances. *Econometric Reviews* 11: 143–172.

Bougerol, P. 1993. Kalman Filtering with Random Coefficients and Contractions. *SIAM Journal on Control and Optimization* 31: 942–959.

[15] The theory is also limited to light tails. Arvanitis and Anyfantaki (2020) show that when the squared innovation has infinite variance and lies in the domain of attraction of a stable law with index in $(1, 2]$, the Gaussian stable QMLE of EGARCH(1,1) converges at a rate slower than $\sqrt{T}$, to a non-Gaussian stable law when the index is below two.

Cont, R. 2001. Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues. *Quantitative Finance* 1: 223–236.

Engle, R. F. 1982. Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. *Econometrica* 50: 987–1007.

Engle, R. 2001. GARCH 101: The Use of ARCH/GARCH Models in Applied Econometrics. *Journal of Economic Perspectives* 15: 157–168.

Engle, R. F., and V. K. Ng. 1993. Measuring and Testing the Impact of News on Volatility. *Journal of Finance* 48: 1749–1778.

Francq, C., and J.-M. Zakoïan. 2004. Maximum Likelihood Estimation of Pure GARCH and ARMA-GARCH Processes. *Bernoulli* 10: 605–637.

Hafner, C. M., and D. Kyriakopoulou. 2021. Exponential-Type GARCH Models with Linear-in-Variance Risk Premium. *Journal of Business & Economic Statistics* 39: 589–603.

Hafner, C. M., and O. Linton. 2017. An Almost Closed Form Estimator for the EGARCH Model. *Econometric Theory* 33: 1013–1038.

Hamilton, J. D. 1994. *Time Series Analysis*. Princeton, NJ: Princeton University Press.

Ling, S., and M. McAleer. 2003. Asymptotic Theory for a Vector ARMA-GARCH Model. *Econometric Theory* 19: 280–310.

Ling, S., and K. Zhu. 2022. Self-Weighted LSE and Residual-Based QMLE of ARMA-GARCH Models. *Journal of Risk and Financial Management* 15: 90.

Meitz, M., and P. Saikkonen. 2011. Parameter Estimation in Nonlinear AR-GARCH Models. *Econometric Theory* 27: 1236–1278.

Nelson, D. B. 1991. Conditional Heteroskedasticity in Asset Returns: A New Approach. *Econometrica* 59: 347–370.

Newey, W. K. 1984. A Method of Moments Interpretation of Sequential Estimators. *Economics Letters* 14: 201–206.

Newey, W. K., and D. McFadden. 1994. Large Sample Estimation and Hypothesis Testing. In R. F. Engle and D. L. McFadden (eds.), *Handbook of Econometrics*, vol. 4, 2111–2245. Amsterdam: Elsevier.

Patton, A. J. 2011. Volatility Forecast Comparison Using Imperfect Volatility Proxies. *Journal of Econometrics* 160: 246–256.

Straumann, D., and T. Mikosch. 2006. Quasi-Maximum-Likelihood Estimation in Conditionally Heteroscedastic Time Series: A Stochastic Recurrence Equations Approach. *Annals of Statistics* 34: 2449–2495.

Tsay, R. S. 2010. *Analysis of Financial Time Series*, 3rd ed. Hoboken, NJ: Wiley.

White, H. 1982. Maximum Likelihood Estimation of Misspecified Models. *Econometrica* 50: 1–25.

Wintenberger, O. 2013. Continuous Invertibility and Stable QML Estimation of the EGARCH(1,1) Model. *Scandinavian Journal of Statistics* 40: 846–867.